

Hometask - 06

Student Name: Reyansh Arora
 Branch: CSE - AIML
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 Management System

UID: 24BAI70273
 Section/Group: 24_AIT_KRG-G1-A
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[A.]

You are given three tables: *Students*, *Friends* and *Packages*. *Students* contains two columns: *ID* and *Name*. *Friends* contains two columns: *ID* and *Friend_ID* (*ID* of the ONLY best friend). *Packages* contains two columns: *ID* and *Salary* (offered salary in \$ thousands per month).

Write a query to output the names of those students whose best friends got offered a higher salary than them. Names must be ordered by the salary amount offered to the best friends. It is guaranteed that no two students got same salary offer.

<i>Column</i>	<i>Type</i>
<i>ID</i>	<i>Integer</i>
<i>Name</i>	<i>String</i>

Students

<i>Column</i>	<i>Type</i>
<i>ID</i>	<i>Integer</i>
<i>Friend_ID</i>	<i>Integer</i>

Friends

<i>Column</i>	<i>Type</i>
<i>ID</i>	<i>Integer</i>
<i>Salary</i>	<i>Float</i>

Packages

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JOIN Friends f ON s.id = f.id
JOIN Packages p1 ON s.id = p1.id
JOIN Packages p2 ON f.friend_id = p2.id
WHERE p2.salary > p1.salary
ORDER BY p2.salary;

```

[B.]

You are given a table, *Functions*, containing two columns: *X* and *Y*

Two pairs (X_1, Y_1) and (X_2, Y_2) are said to be *symmetric pairs* if $X_1 = Y_2$ and $X_2 = Y_1$.

Write a query to output all such *symmetric pairs* in ascending order by the value of *X*. List the rows such that $X_1 \leq Y_1$

Column	Type
X	Integer
Y	Integer

ANS:

```

SELECT f1.x, f1.y
FROM Functions f1
JOIN Functions f2
  ON f1.x = f2.y
  AND f1.y = f2.x
GROUP BY f1.x, f1.y
HAVING COUNT(*) > 1 OR f1.x < f1.y
ORDER BY f1.x;

```